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Impact of Educational Development Programs on Teaching Self-Efficacy in Graduate Students: A Systematic Literature Review

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Abstract

This study reviews 33 studies to analyze the impact of educational development programs on the teaching self-efficacy of graduate students. It primarily employs quantitative and mixed research methods. Three original scales were predominantly used for surveys, with many studies modifying these scales. Analysis revealed that irrespective of the duration, educational development programs led to an improvement in participants' teaching self-efficacy. The direct experiential learning provided by microteaching had the most significant effect on enhancing teaching self-efficacy. However, some studies indicated no significant difference in participants' teaching self-efficacy before and after program participation, reasons for which were analyzed in this study.

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1. Introduction

Graduate training has primarily focused on the development of research scholars, however, recently teaching competencies are being increasingly recognised as a necessary component of graduate education [1]. To meet the professional development needs of graduate students, a significant increase in educational development programs has occurred, with scholars increasingly recognizing the need to assess the impacts of such initiatives [2]. Many universities provide general educational development programs and some form of specialized training for Graduate Teaching Assistants (GTAs), however, the validation of the effects of these educational development programs within higher education remains diverse and fragmented.

Substantial evidence indicates a significant relationship between teaching effectiveness and self-efficacy [3], with this relationship being reciprocal. Despite the urgent need for effective graduate students induction, limited information is available regarding how developmental opportunities enhance their teaching effectiveness [4].

This study is grounded in Social Cognitive Theory (SCT) and adopts the widely recognized theoretical framework of Albert Bandura [5], where instructors' beliefs in their self-efficacy and their behaviors are typically analyzed. As a central concept of SCT, Bandura defines self-efficacy as "the belief in one's capabilities to organize and execute the courses of action required to manage prospective situations." In teaching contexts, such beliefs can influence course planning, decision-making, the adoption of certain teaching practices, and subsequently affect relationships with students, their achievements, and success. Many instructors require support to better understand and regulate their thoughts and behaviors, including selecting personal goals, perceiving sources of stress, setting expectations for student behavior and performance, and managing increased teaching, research, and service workloads.

Therefore, this study aims to synthesize and integrate existing literature on the use of teaching self-efficacy as a measure for validating the effects of educational development programs. This includes identifying common themes and gaps in the literature and indicating new directions for research development. This approach contributes to the academic field by mapping out an evidence-based understanding of the impact of educational development programs and aids in the formulation of strategies to enhance the teaching capabilities of graduate teaching assistants.

2. Methodology

2.1. Research Questions

This study explores the role of educational development programs in enhancing teaching self-efficacy among graduate students, surveying studies published since 1997. We addressed the following research questions to understand the depth of this transformation:

- RQ1. What research methods are used in studies on teaching self-efficacy in graduate student educational development programs?
- RQ2. What are the types and structures of measures of teaching self-efficacy for graduate students?
- RQ3. Do graduate student educational development programs affect teaching self-efficacy? If yes, what effect does the program have on teaching self-efficacy and what factors in the program have an effect?

We identified and categorized the methods scholars use to study teaching self-efficacy (RQ1). Thereafter, we detailed the various instruments, particularly surveys, used for this purpose (RQ2). Finally, we evaluated how these educational programs influence teaching self-efficacy, focusing on the most impactful program elements (RQ3).

2.2. Review Process

To conduct an objective review, we adhered to stringent inclusion and exclusion criteria. These criteria, detailed in Table 1, required that studies be peer-reviewed, written in English, and published in the last 30 years. They needed to specifically focus on teaching self-efficacy and include both in-service and pre-service faculty, as well as graduate students and postdoctoral scholars. We excluded any studies that were not peer-reviewed, not written in English,

published before 1993, not focused on teaching self-efficacy, not related to higher education, or were primarily concerned with developing scales rather than program evaluation.

Table 1. Inclusion and exclusion criteria.

	Inclusion Criteria	Exclusion Criteria
Reason 1	Peer-reviewed studies	Studies proceedings without peer review
Reason 2	Studies written in English	Studies written in languages other than English
Reason 3	Studies published in the last 30 years from 1993 to 2023	Studies published before 1993
Reason 4	Studies focusing on teaching self-efficacy	Measures of teaching self-efficacy that include aspects beyond teaching
Reason 5	The research target group comprised in-service and pre-service faculty members (graduate students, postdocs, and faculty/staff members)	Studies targeting pre-service teacher outside of higher education or only in-service teachers
Reason 6	Practical studies that aimed at evaluating the effectiveness of a program	Studies where the chief focus was not on program effectiveness evaluation, but on the development of scales or models

Following the PRISMA 2020 guidelines [6], our review process used a comprehensive checklist and flowchart, presented in Figure 1, to maintain transparency and consistency. We implemented a detailed search strategy across several databases, using terms specifically selected to find relevant studies on educational development programs for graduate students. The search terms applied to all databases were "((Preparing) Future Faculty Program) OR (University Teacher Preparation Program) OR (Undergraduate Teacher Preparation Program) OR (GTA) AND (Teaching Self-Efficacy)" with the inclusion of a "peer-reviewed" filter. This approach produced varied results: ERIC (14), PsycINFO (7), Google Scholar (369), and Web of Science (17).

We carefully screened 247 studies for relevance after removing duplicates. Each study was independently evaluated by three authors using the pre-determined criteria. Initially, we discussed and resolved discrepancies and reached an agreement in 88.3% of the cases. The remaining 29 studies required further evaluation.

Following the structured process illustrated in Figure 1, we meticulously performed title and abstract screenings of the identified records. Articles that passed this stage were subjected to full-text review for a detailed eligibility assessment. Our collaborative effort resulted in the selection of 33 studies that robustly met our criteria and contributed valuable insights into the impact of educational development programs on teaching self-efficacy among graduate students.

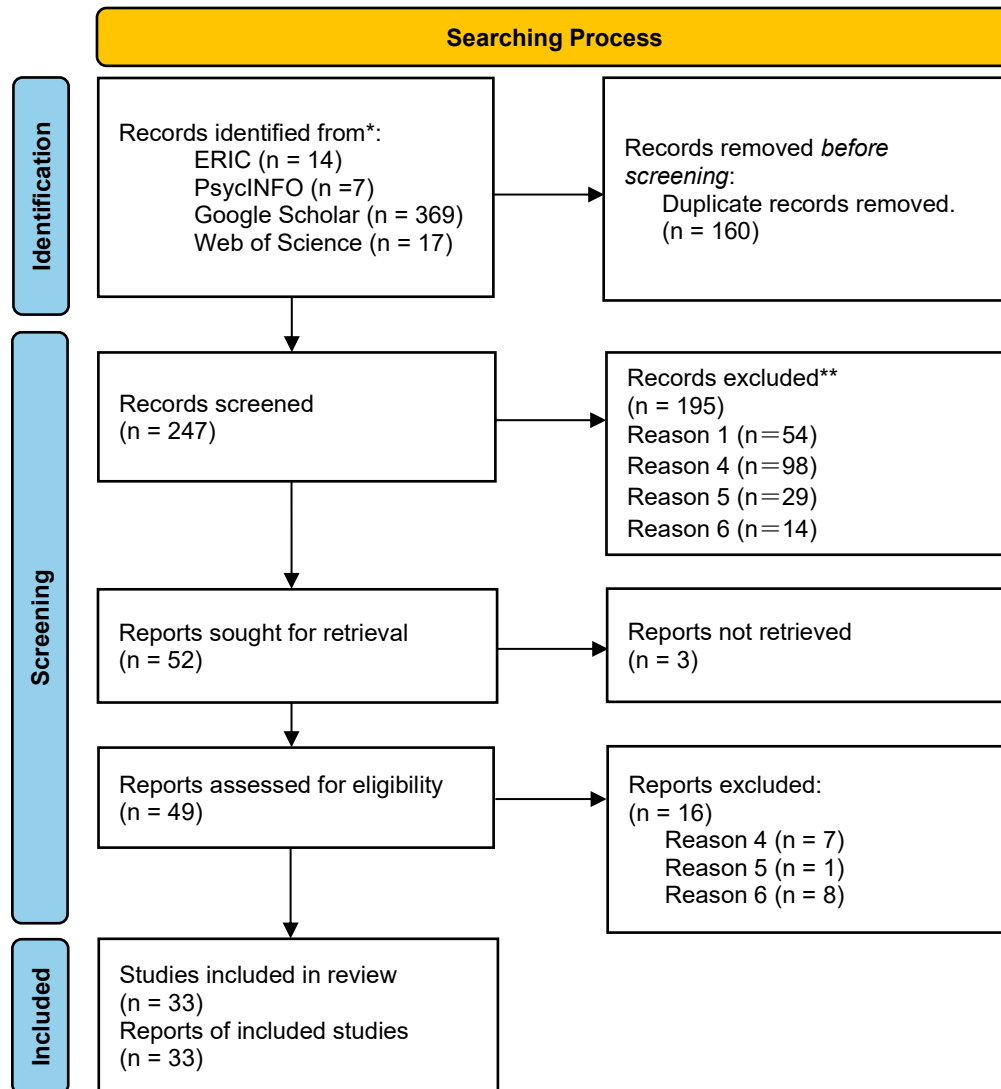


Fig. 1. PRISMA flow chart summarizing the review process.

3. Results

According to the criteria of the literature review, a total of 33 studies [7-39] were selected. These studies investigated graduate student educational development programs conducted in higher education settings, with the target population primarily being graduate students. Regarding the specialties of the graduate student educational development programs and the target population of the training projects, the focus was primarily on STEM (Science, Technology, Engineering, and Mathematics). STEM education aims to cultivate students' skills in science, technology, engineering, and mathematics to address the challenges and demands of modern society.

3.1. RQ1. What research methods are used in studies on teaching self-efficacy in graduate student educational development programs?

As presented in Table 2, among the 33 studies examined, 17 studies employed a mixed methods approach, combining qualitative and quantitative research methodologies. Apart from the mixed methods, 12 studies employed quantitative research methods, whereas 4 studies employed qualitative research methods.

Among the various survey methods used, questionnaire surveys emerged as the most frequently employed research tool. Furthermore, a majority of the studies employed either questionnaire surveys or interviews. This suggests a certain level of representativeness in the selected research methodologies. The subsequent sections provide a detailed overview of the usage patterns of these methods.

Table 2. Research methods in reviewed studies (N = 33).

Quantitative (12)	Mixed methods (17)	Qualitative (4)
• Questionnaires (12)	• Questionnaires (17) • Interviews (10) • Open-ended questions (6) • Observation (3) • Written reflection (2)	• Interviews (3) • Observation (2) • Open-ended questions (1) • Documents (1)

• Questionnaires

All studies employing mixed and quantitative research methods used questionnaire surveys for data collection. In some studies, scholars conducted multiple rounds of questionnaire surveys to understand the changes in the responses of participants of graduate student educational development programs, particularly comparing their responses before and after participating in the program. To collect comprehensive data from program participants, usually questionnaires include questions in multiple areas. In addition to demographic questions and items from the Teaching Self-Efficacy Scale, various other questions were employed to measure other aspects, such as satisfaction with the program. For instance, to comprehensively evaluate the effects of the Facilitated Learning to Advance Geriatrics(FLAG) Program, participants completed a brief demographic survey and the College Teaching Self-Efficacy Scale (CTSES) [23]. Additionally, they completed the CTSES and an assessment of satisfaction with the program at the conclusion of the workshops. Moreover, some questionnaires investigated psychological aspects such as anxiety [17]. The scholars adapted and modified the generalized version of the Confidence and Anxiety scale to assess self-efficacy and anxiety in teaching at pre-test and post-test stages.

Regarding questionnaire format, questions scored on a Likert scale were the most popular. For example, concerning training objectives, a five-point Likert scale was employed to rate each objective, ranging from "Strongly Agree" (5 points) to "Strongly Disagree" (1 point) [37]. Furthermore, in another study, the first section of the digital questionnaire comprised a set of 18 Likert-style items regarding self-efficacy [26].

The specific types of scales and questions related to teaching self-efficacy used in the questionnaire survey is detailed in the subsequent section addressing Research Question 2.

• Interviews

Focus group interviews are the primary method of data collection in qualitative research because they effectively gather rich and detailed insights. Using this method, scholars can acquire in-depth understanding of participants' perspectives, experiences, and evaluations of various programs. Typically conducted as follow-up interviews after the conclusion of a program, these interviews aim to track changes in participants' viewpoints or collect more nuanced data.

For instance, in one study, scholars conducted a series of six focus group interviews, each accommodating 2 to 6 participants [19]. It aimed to explore participants' perceptions of the effectiveness and relevance of the graduate student educational development programs in which they participated. The inquiry comprised two aspects: first, participants were asked to reflect on their overall perceptions of the program; second, discussions revolved around the applicability of the teaching strategies they learned in their own teaching practices. This dual approach allowed scholars to acquire a comprehensive understanding of the direct and long-term impacts of the training program on participants' teaching methods. Additionally, another study focused particularly on participants' experiences in the

FLAG program [23]. In this study, scholars meticulously designed interview questions to examine the specific impact of participating in the FLAG program on participants' teaching skills.

3.2. RQ2. What are the types and structures of measures of teaching self-efficacy for graduate students?

The results to Research Question 2 provide an overview of the teaching self-efficacy scale used to measure the effectiveness of graduate student educational development programs. The scales used in several studies in the reviewed literature are listed in Table 3.

Table 3. Teaching self-efficacy scales commonly used in reviewed studies.

Scale name [Original citation]	Version	Context	Citations in this review
GTA Teaching Self-Efficacy Scale [32]	Original version (DeChenne <i>et al.</i> , 2012)	The scale is designed to survey GTAs in STEM and consists of two factors. • Instructional strategies (7 items) • Learning environment (11 items)	[12], [16], [26], [27], [30], [32], [33]
	15 items version (Ke <i>et al.</i> , 2019; 2021)	For unspecified reasons, these studies used a reduced 15 item version aimed at GTAs in STEM.	[22], [28]
	11 items version (Fong <i>et al.</i> , 2019)	To reduce survey fatigue and increase completion rates, a shortened version of the 11 items retaining the two factors was developed and expanded the target audience to GTAs across all disciplines. • Instructional strategies (5 items) • Learning environment (6 items)	[34]
Teaching Assistant Self-Efficacy Scale [40, 41] (TASE) scale	Original version (Boman, 2008; 2013)	The original version comprises an item on overall confidence as a TA and the following three subscales. • Interaction (15 items) • Written (12 items) • Improvement (4 items)	[7], [9], [19]
	15 teaching parameters (Troop <i>et al.</i> , 2015)	Adapted Boman's to a 10-point scale with 15 items.	[13]
	26 items version (Sozer <i>et al.</i> , 2019)	Factor analysis revisited three subscales. • Delivery (11 items) • Interaction (10 items) • Assessment (5 items)	[20]
Teachers' Sense of Efficacy Scale [42] (TSES)	12 items short forms (Tschannen-Moran <i>et al.</i> , 2001)	Both the short and long forms comprise the following three factors. • Instructional strategies (4 or 8 items) • Classroom management (4 or 8 items) • Student engagement (4 or 8 items)	[24], [25]
	24 items long forms (Tschannen-Moran <i>et al.</i> , 2001)		

The STEM GTA Teaching Self-Efficacy Scale (STEM GTA-TSES), developed by DeChenne, Enochs, and Needham in 2012, is among the most widely used scales [12, 16, 22, 26-28, 30, 33, 34]. This scale evaluates the

teaching self-efficacy of teaching assistants in STEM disciplines. It employs a 5-point Likert scale with subscales for "Instructional Strategies" and "Learning Environment." The original version comprises 18 items, although versions with 15 [16,22] and 11 [28] items also exist. The 11-item shortened version developed by Fong uses a 4-point Likert scale and retains the same subscales while being tested on a diverse group of graduate students beyond STEM [28].

Another scale, the Teaching Assistant Self-Efficacy (TASE) Scale, was developed by Boman [40, 41]. The original 5-point Likert scale focuses on overall confidence as a TA, with items on confidence in interacting with students and teaching (15 items), confidence in class preparation and course-related writing (12 items), and confidence in enhancing teaching abilities (4 items) [7, 9, 19]. Subsequent studies have adapted it into a 10-point Likert scale with 15 teaching parameters [13] and a 26-item version reorganized into three subscales: Delivery (11 items), Interaction (10 items), and Assessment (5 items) through factor analysis [20].

Furthermore, the Teachers' Sense of Efficacy Scale (TSES), developed by Tschannen-Moran and Hoy in 2001, comprises subscales for "Instructional Strategies," "Classroom Management," and "Student Engagement," with both a 12-item short form and a 24-item long form developed on a 9-point scale [41]. Originally developed for use in elementary and secondary education contexts, this scale requires adaptation for higher education settings. Modifications may include measuring only the items from "Instructional Strategies" and "Student Engagement" factors [24], excluding items not suitable for higher education [25], or customizing the term "school" to "class" [35].

Several other scales also exist. These scales are invaluable for educators to understand their perceived competencies and play a crucial role in developing educational programs and strategies. Each scale is designed to meet the specific challenges and demands faced by educators in different educational contexts.

3.3. RQ3. Do graduate student educational development programs affect teaching self-efficacy? If yes, what effect does the program have on teaching self-efficacy and what factors in the program have an effect?

When evaluating the effects of educational development programs on graduate students' self-efficacy, the most common methodology is the pre-post t-test analysis. Across both short-term and long-term programs, a significant majority of studies indicate marked improvements in participants' self-efficacy pre- and post-engagement [7, 8, 9, 11, 12, 13, 17, 19, 23, 24, 27, 28, 30, 31, 35, 36, 37, 39]. Higher levels of confidence about teaching are held by participants in more extended programs, spanning several weeks to semesters, compared with those in shorter programs lasting a few hours to three days [9].

The most effective method to boost self-efficacy involves practical application, such as conducting teaching demonstrations [8] or engaging in micro-teaching [11, 19, 24, 31]. Additionally, the support of peers and the practice of reflective teaching contribute significantly to enhancing self-efficacy [13, 21, 24]. Specifically, teaching activities that increased participants' self-efficacy included drafting teaching-related materials [36], classroom management and interaction [19, 20, 27, 35], as well as implementing teaching strategies [9, 13, 36]. Moreover, micro-teaching ensures practice within a safe environment, allowing participants to diminish their communication fears [13, 15, 19, 31, 39] and venture into new techniques [19], thereby furthering the promotion of student-centered teaching approaches [24, 31, 39] across both STEM and humanities disciplines [37].

Furthermore, experienced graduate students typically possess higher levels of self-efficacy [20,38], aligning with Bandura's theory of self-efficacy. Graduate students lacking prior experience achieve more significant self-efficacy improvements through training programs [17]. Moreover, surveys indicate that the teaching experience of a graduate student [21,34] or the mode of simulation used—VR or reality [22,28]—does not significantly impact the development of teaching self-efficacy.

However, a subset of literature suggests no significant differences in self-efficacy pre- and post-training program participation [16,20,27,35]. In extreme cases, where pre-program self-efficacy is either too high or too low, participants' overconfidence or lack thereof may render the program ineffective at inducing meaningful changes in self-efficacy. Moreover, some participants view simulated teaching as lacking realism, potentially affecting the transfer of self-efficacy to actual classroom settings [13].

4. Discussion

This review underscores the significance of teaching self-efficacy within graduate student educational development programs, identifying it as a crucial yet underexplored area. The insight derived from Research Question 1 is that studies on measuring the validation of the effectiveness of educational development programs have unfolded primarily through insights derived from various methods, primarily quantitative and mixed research methods. The overwhelming preference for questionnaire scales in educational psychology research can be attributed to their expertise in psychometrics, providing a precise, adaptable and scientifically rigorous method to measure complex psychological constructs such as teaching self-efficacy. These scales enable standardized, objective assessments across populations and contexts, reflecting the discipline's emphasis on validated, reliable measurement.

A widespread demonstration of increased rigor in the development and selection of specific items is evident in studies employing questionnaires. In the analyses conducted in response to Research Question 2 reveals that despite the existence of three widely cited original scales for measuring teaching self-efficacy, there is movement toward the development and use of more "customized" research tools tailored to different disciplinary domains and scales. These tools promise to offer more nuanced insights than the broader "general" findings currently available, potentially enriching the field significantly and yielding more diagnostic outcomes.

Mastery experiences, as outlined by Bandura [5], are recognized as the most potent source of self-efficacy. The analysis of Research Question 3 verifies that accumulating experiences in micro-teaching activities, which provide a safe environment for trying new techniques, may impose less psychological pressure than real classroom situations while still enhancing teaching self-efficacy effectively for graduate students.

However, when the analysis of Research Question 3 revealed that the sense of teaching self-efficacy did not produce an intentional difference before and after the participation in the project, it was analyzed that this could be owing to three circumstances. When confronting the extremities of teaching self-efficacy, whether exceedingly high or low, the issues may stem from the measurement scales' insensitivity, indicating a limitation of the instruments used. Moreover, societal disapproval of anxiety and lack of confidence [43] may lead educators to overestimate their confidence levels and potentially result in a ceiling effect for high teaching self-efficacy. Conversely, graduate students with initially low teaching self-efficacy may find that graduate student educational development programs are insufficient to elevate their levels. This highlights the necessity for customized interventions, considering the diverse initial levels of teaching self-efficacy and interest among participants, to bolster their teaching effectiveness and confidence meaningfully. At a more granular level, studies into how variations in discipline, time, and program design collectively influence graduate students could inform the development of educational development programs tailored to specific local needs.

Additionally, although teaching self-efficacy beliefs can influence graduate students' choice of teaching methods, encouraging more student-centered strategies that facilitate learning and studies regarding how graduate students' beliefs affect their teaching practices remains scarce. Considering that teaching self-efficacy ultimately measures graduate students' confidence in their teaching capabilities, objective evaluations of their teaching practices should also be considered within educational development programs as a measure of success.

In the contemporary higher education landscape, universities are preparing for a post-pandemic future as new challenges emerge, such as the transformation of teaching modalities through online activities, the adoption of new technologies, and the evolving dynamics of interactions between colleagues and students. For instance, VR-based teaching simulations have been suggested as effective alternatives to enhance teaching self-efficacy through dynamic and immersive experiences. Moreover, institutional support for both students and GTAs is crucial in navigating these challenges, making the study of graduate student teaching self-efficacy more relevant than ever.

5. Conclusion

This study conducts a systematic review of 33 studies since 1993, providing a comprehensive overview of studies on the impact of various educational development programs on the teaching self-efficacy of future teachers. Most studies employ quantitative and mixed research methods. Regarding research methodologies, the majority of studies employ questionnaire surveys, with scholars administering surveys to measure participants' teaching self-efficacy. Consequently, this study summarizes the scales used to measure teaching self-efficacy across various studies and

elaborates on the factors and constructs of commonly used scales. Additionally, the study summarizes and discusses the specific effects of educational development programs on participants' teaching self-efficacy and the associated influencing factors.

To minimize errors and enhance validity, all steps of this study were completed and verified by three individuals. However, this study has certain limitations. This review primarily relies on quantitative and mixed method studies used in the literature, which may lead to a bias toward quantitative assessment of teaching self-efficacy, while lacking a deep understanding of individual experiences and contextual factors. Furthermore, this review did not provide detailed discussion on the content of educational development programs. Programs offered at different times or with different content may have varying effects.

Future research is required for incorporating more qualitative research methods, such as observations, and case studies, to acquire a more comprehensive understanding of the changes in teaching self-efficacy among future teachers. Additionally, most existing studies may focus on the short-term impact of educational development programs on teachers' self-efficacy. Future research should implement long-term tracking to examine the effects of these programs on teachers' long-term teaching practices, career development, and changes in self-efficacy.

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